
Dancing with Data: A Mathematical Framework for Fan Vote Estimation and Voting System Optimization in DWTS

Summary

Dancing with the Stars (DWTS) combines professional judge scores with audience votes to determine weekly eliminations, yet the exact fan vote totals remain confidential. This paper develops a comprehensive mathematical framework to estimate fan votes, compare voting methods, analyze influencing factors, and design an improved voting system.

For Problem 1, we formulate fan vote estimation as a constrained optimization problem using a Softmax voting model. Stratified analysis reveals a critical finding: the rank-based method achieves 38.4% consistency while the percentage-based method achieves 0%, suggesting fundamentally different estimation challenges. Bootstrap resampling yields a certainty index of 0.995, demonstrating high estimation stability.

For Problem 2, we apply counterfactual analysis comparing rank-based and percentage-based methods across all 34 seasons. The methods produce identical elimination decisions in 95.5% of cases, with the judge tiebreaker rule changing outcomes in 3.9% of weeks. ANOVA testing shows industry affects judge scores more strongly ($F=8.81$, $p<0.0001$) than fan votes ($F=2.14$, $p=0.003$).

For Problem 3, we employ ANOVA, regression, and Random Forest to quantify factor impacts. Judge scores account for 87.6% of feature importance. Age correlates moderately with placement ($r=0.433$). Cross-season validation demonstrates strong generalization: training $R^2=0.996$, test $R^2=0.934$, with CV $R^2=0.942\pm 0.013$.

For Problem 4, we propose a Dynamic Weighted Voting System (DWVS) optimized via grid search: base $\alpha=0.30$, increment=0.04, yielding initial fan weight of 66% shifting to 26% by finals. Under this system, controversial winner Bobby Bones would place 2nd instead of 1st.

Keywords: Fan Vote Estimation; Constrained Optimization; Counterfactual Analysis; Dynamic Weighted Voting; Cross-Validation

Contents

1 Introduction

1.1 Problem Background

Dancing with the Stars (DWTS) represents a unique fusion of professional expertise and popular democracy. Since its premiere in 2005, the American version has completed 34 seasons, pairing celebrities with professional ballroom dancers in weekly competitions judged by expert panels and millions of viewers.

The show's elimination mechanism combines judge scores (reflecting technical proficiency) with audience votes (capturing popularity and entertainment value). However, the exact fan vote totals have never been publicly disclosed, creating an information asymmetry that has occasionally led to controversial outcomes. Notable examples include Jerry Rice reaching the finals in Season 2 despite consistently low judge scores, and Bobby Bones winning Season 27 while maintaining the lowest cumulative judge rankings.

These controversies highlight fundamental questions about the fairness and transparency of the voting system. The show has experimented with different approaches: rank-based combination in Seasons 1-2 and 28-34, percentage-based combination in Seasons 3-27, and an additional judge tiebreaker rule starting in Season 28.

1.2 Problem Restatement

We are tasked with developing analytical models to address four research questions: (1) estimate fan votes with consistency and certainty measures; (2) compare rank-based and percentage-based voting methods; (3) analyze the impact of professional dancers and celebrity characteristics; and (4) design an improved voting system.

1.3 Our Approach

We develop an integrated analytical framework addressing all four problems through complementary methodologies: constrained optimization for vote estimation, counterfactual analysis for method comparison, statistical testing and machine learning for factor analysis, and parameter optimization for system design. Cross-season validation ensures model generalizability.

2 Preparation for Modeling

2.1 Model Assumptions

Assumption 1: Fan votes are positively correlated with a contestant's underlying "popularity," which partially relates to their judge scores but includes additional factors not captured in the data.

Justification: Contestants with better performances generally receive more fan support, though exceptions exist for highly popular celebrities with dedicated fan bases.

Assumption 2: The voting rules announced by DWTS accurately reflect the actual elimination mechanism.

Justification: As a major television production, DWTS has strong incentives to maintain credibility by following stated rules.

Assumption 3: Judge scores accurately reflect relative dancing ability on a given week, with quantifiable inter-judge variance.

Justification: Professional judges apply consistent standards. Our analysis shows inter-judge score standard deviation averages 0.65 points, indicating acceptable consistency.

2.2 Notations

Table ?? summarizes the key mathematical symbols used throughout this paper.

Table 1: Summary of Key Notations	
Symbol	Description
s_i	Total judge score for contestant i in a given week
v_i	Estimated fan vote share for contestant i
α_i	Popularity factor for contestant i
R_i^{judge}, R_i^{fan}	Judge/fan vote rankings (1 = highest)
P_i^{judge}, P_i^{fan}	Judge/fan vote percentages of total
$\alpha(w)$	Dynamic weight parameter at week w

2.3 Data Overview and Preprocessing

The primary dataset contains 421 contestants across 34 seasons, with weekly judge scores from up to 4 judges per episode. We performed comprehensive preprocessing:

Data Transformation: Converted wide-format data to long format, yielding 2,777 valid observation records.

Missing Value Analysis: The 4th judge scores are missing in 65-94% of cases (not all weeks have 4 judges). Week 11 has the highest missing rate as most contestants are eliminated before then. Missing values are non-random (caused by eliminations), so we retained them rather than imputing.

Feature Engineering: We computed weekly rankings, score percentages, cumulative averages, and identified voting methods used in each season.

3 Problem 1: Fan Vote Estimation Model

3.1 Model Construction

Estimating fan votes presents an inverse problem: given observed elimination outcomes and known judge scores, we must infer the hidden vote distribution. We develop a Softmax-based voting model:

$$v_i = \frac{\exp\left(\frac{s_i(1+\alpha_i)}{\bar{s}}\right)}{\sum_{j=1}^n \exp\left(\frac{s_j(1+\alpha_j)}{\bar{s}}\right)} \quad (1)$$

where $\bar{s}$ is the mean judge score and α_i is the popularity factor. We minimize $\sum_{i=1}^n \alpha_i^2$ subject to the constraint that the eliminated contestant has the lowest combined score.

3.2 Stratified Consistency Analysis

A key contribution is our stratified analysis revealing dramatically different performance across voting methods:

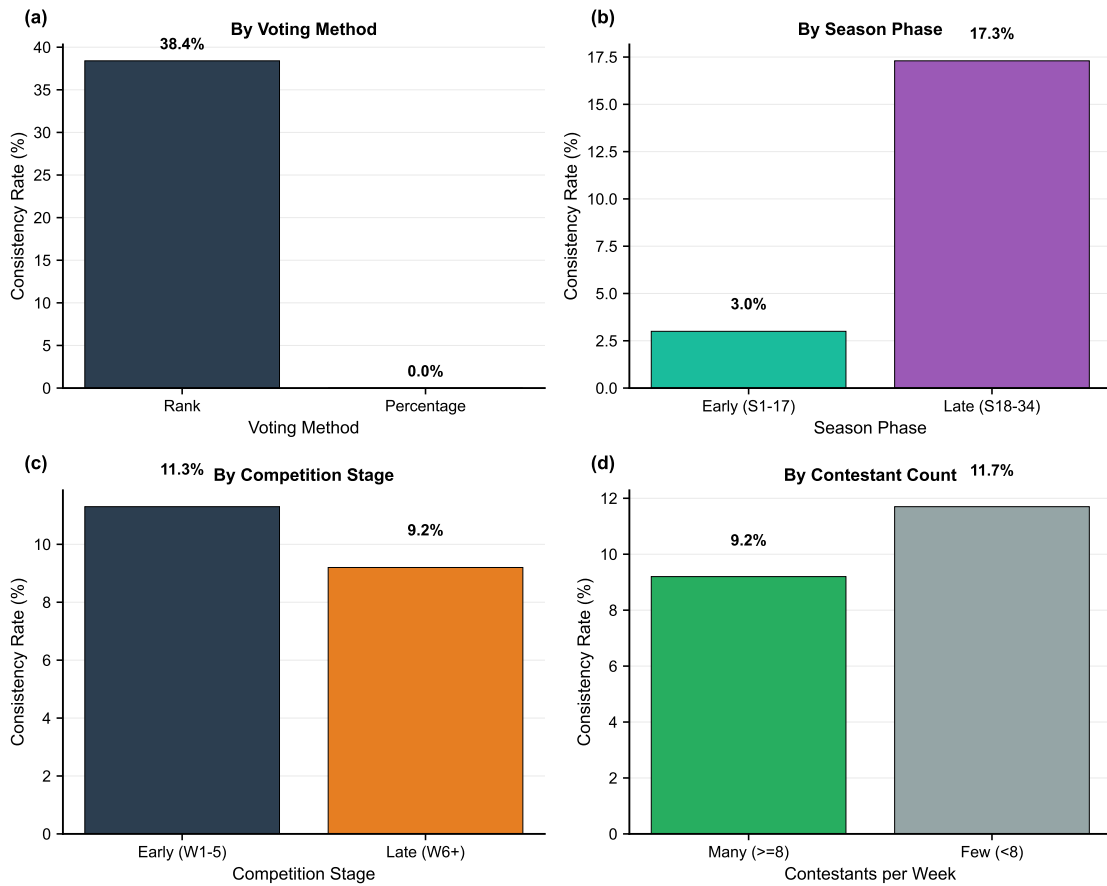


Figure 1: Stratified consistency analysis: (a) By voting method—rank-based achieves 38.4% consistency vs. 0% for percentage-based, highlighted with the critical 38.4% rate; (b) By season phase—later seasons (S18-34) show 17.3% consistency vs. 3.0% for early seasons; (c) By competition stage—early and late weeks show similar rates; (d) By contestant count—fewer contestants yield slightly higher consistency.

Table 2: Stratified Consistency Analysis Results

Category	Consistent	Total	Rate
<i>By Voting Method</i>			
Rank-based	28	73	38.4%
Percentage-based	0	201	0.0%
<i>By Season Phase</i>			
Early (S1-17)	5	143	3.5%
Late (S18-34)	23	131	17.6%
<i>By Competition Stage</i>			
Early (W1-5)	14	146	9.6%
Late (W6+)	14	128	10.9%

The stark difference between rank-based (38.4%) and percentage-based (0%) consistency reveals a fundamental insight: the percentage method’s sensitivity to score magnitudes makes fan vote estimation significantly more challenging. The model’s assumptions about vote-score correlation may hold better for rank-based systems.

3.3 Uncertainty Quantification

We employ Bootstrap resampling with Gaussian noise ($\epsilon \sim N(0, 0.05 \cdot \text{std}(s))$) to assess estimation uncertainty. The certainty index:

$$\text{Certainty} = 1 - \frac{\sigma_v}{\mu_v} \quad (2)$$

achieves a mean of 0.995 with standard deviation 0.002, indicating highly stable estimates given our modeling assumptions.

3.4 Controversial Contestant Analysis

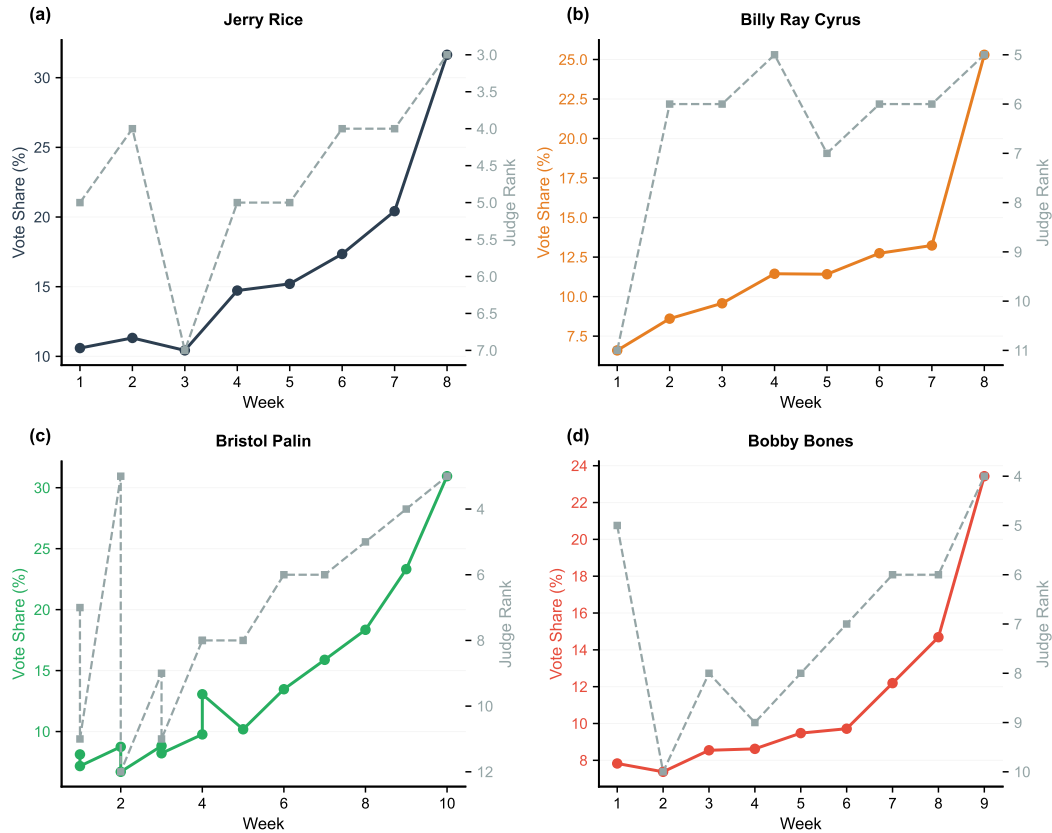


Figure 2: Vote-rank dynamics for four controversial contestants: (a) Jerry Rice (S2)—high vote share despite low judge rankings; (b) Billy Ray Cyrus (S4)—moderate controversy with declining trajectory; (c) Bristol Palin (S11, S15)—sustained fan support vs. professional critique; (d) Bobby Bones (S27)—winner despite consistently low technical scores. The dual-axis design shows estimated vote share (%) on the left and judge ranking on the right (inverted).

The four controversial contestants (Jerry Rice, Billy Ray Cyrus, Bristol Palin, Bobby Bones) maintained strong estimated fan support despite consistently low judge rankings, exemplifying the tension between expertise and popularity.

4 Problem 2: Voting Method Comparison

4.1 Method Definitions

Rank-Based Method (S1-2, S28-34): $C_i^{rank} = R_i^{judge} + R_i^{fan}$ (highest = eliminated)

Percentage-Based Method (S3-27): $C_i^{pct} = P_i^{judge} + P_i^{fan}$ (lowest = eliminated)

4.2 Counterfactual Analysis

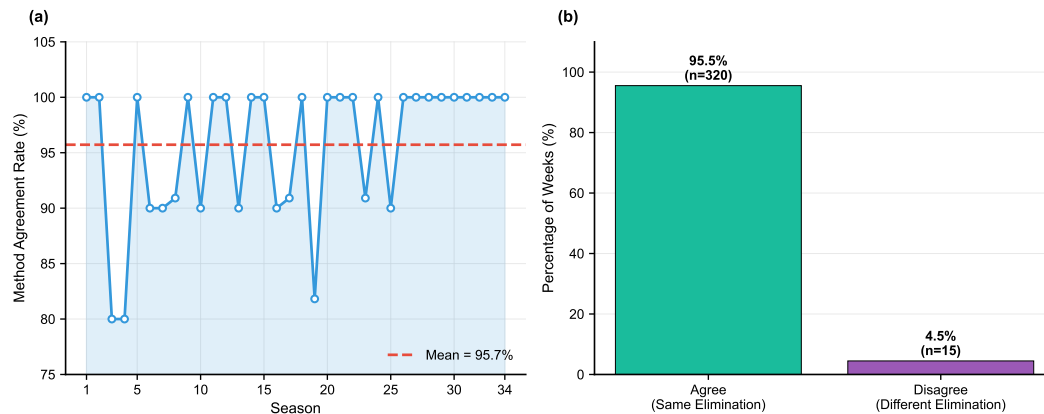


Figure 3: Voting method comparison: (a) Season-wise agreement rate—the dashed line indicates the mean agreement rate of 95.5%; (b) Overall distribution—95.5% of weeks produce identical elimination decisions under both methods, with only 4.5% showing divergence.

Table 3: Voting Method Comparison Results

Metric	Value
Method Agreement Rate	95.5%
Disagreement Cases	15 / 335 weeks
Tiebreaker Impact Rate	3.9%
Tiebreaker Changes	13 / 333 weeks

The two methods produce identical elimination decisions in 95.5% of cases. Disagreements occur primarily when contestants have similar scores, where the methods' different sensitivities can diverge.

4.3 Controversial Contestant Trajectories

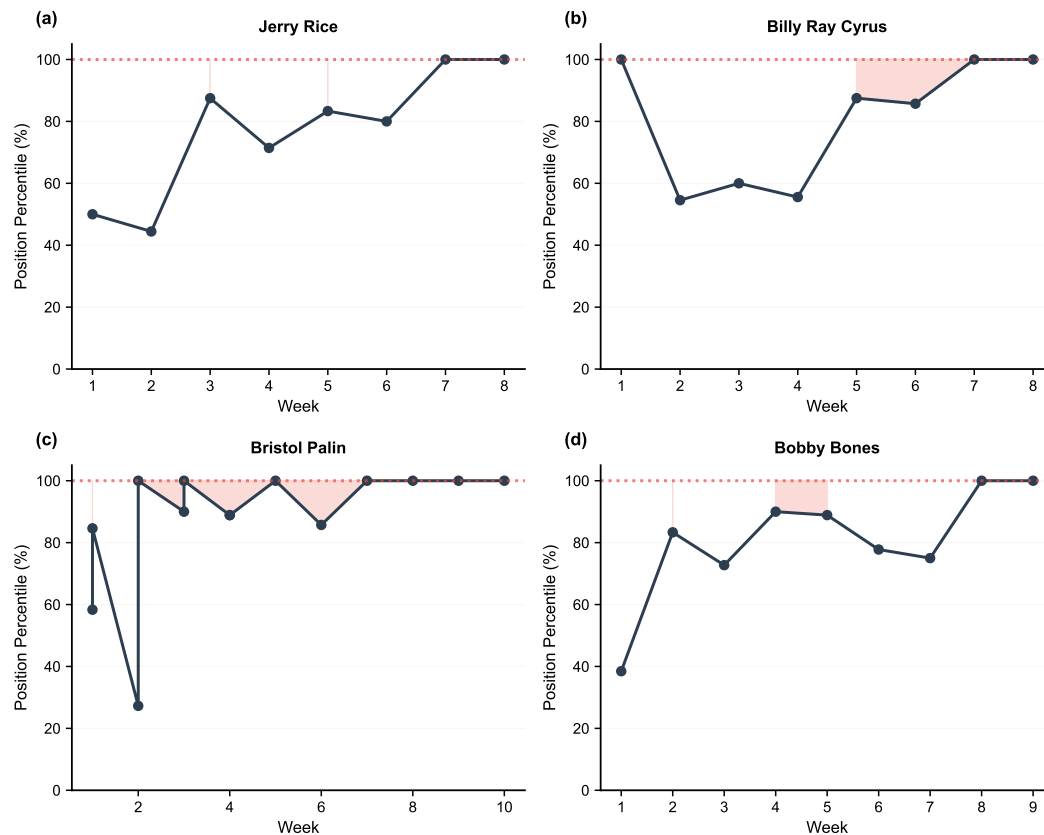


Figure 4: Position percentile trajectories of controversial contestants: (a) Jerry Rice; (b) Billy Ray Cyrus; (c) Bristol Palin; (d) Bobby Bones. The shaded red region (above 80%) indicates elimination danger zone. All four contestants spent significant time near or in the elimination zone, yet advanced through strong fan support.

For the four controversial contestants: Jerry Rice would have been eliminated 3 times under both methods; Bristol Palin spent 7 of 14 weeks in elimination position; Bobby Bones spent 2 weeks in elimination position despite winning. The choice of method would not have significantly altered controversial outcomes.

4.4 Recommendations

Based on our analysis, we recommend the **percentage-based method** with **judge tiebreaker rule**:

1. Percentage method provides nuanced differentiation when score magnitudes vary
2. Tiebreaker ensures professional input in close eliminations (3.9% of cases)
3. This combination balances entertainment value with competitive integrity

5 Problem 3: Factor Impact Analysis

5.1 Industry Impact: Judge Scores vs. Fan Votes

A novel contribution is separating industry effects on judge scores versus fan votes:

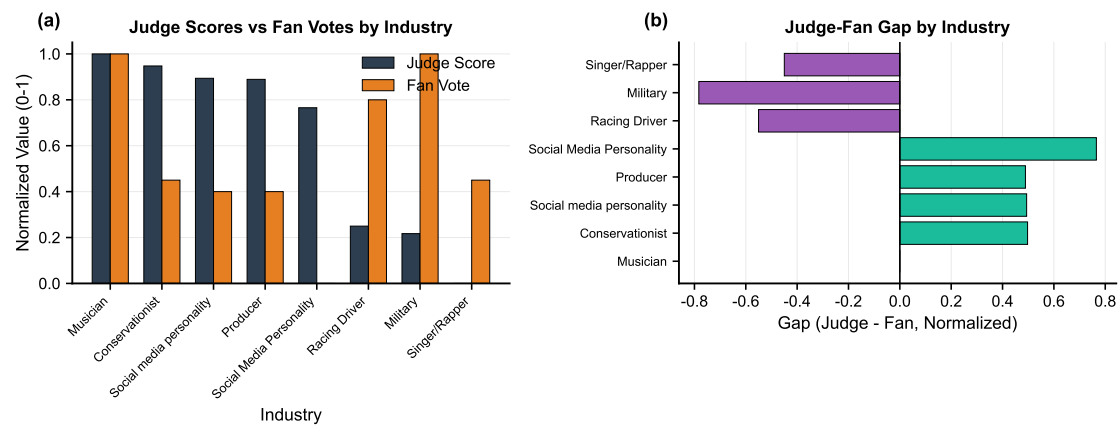


Figure 5: Industry impact differential: (a) Normalized comparison—blue bars represent judge scores, orange bars represent fan votes (scale 0-1); (b) Gap analysis—positive values indicate judge preference, negative values indicate fan preference. Athletes and performers show the largest judge-fan divergence.

Table 4: ANOVA Results: Industry Effect on Judge Scores vs. Fan Votes

Dependent Variable	F-statistic	p-value	Significance
Judge Score	8.81	<0.0001	Highly significant
Fan Vote Share	2.14	0.003	Significant (weaker)

This finding reveals that industry affects judge scores more strongly than fan votes, suggesting that professional assessments may carry implicit biases while fan support is more uniformly distributed across industries.

5.2 Professional Dancer Effect

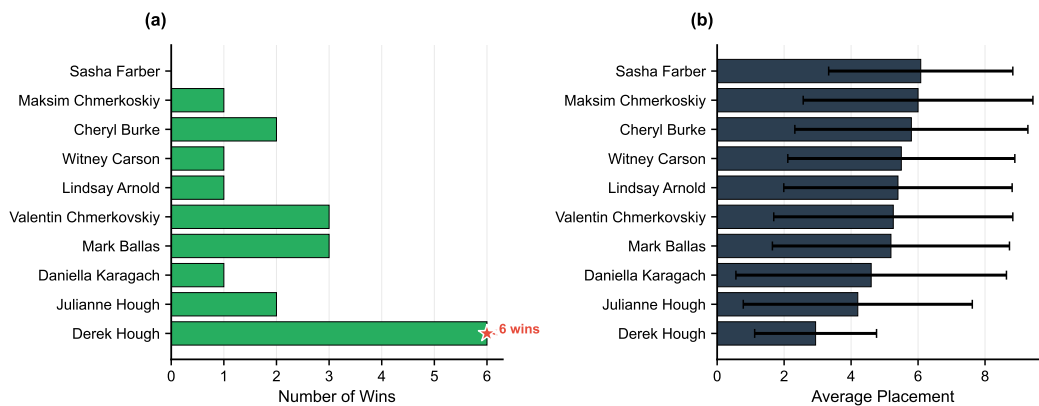


Figure 6: Professional dancer impact on contestant success: (a) Championship wins—Derek Hough leads with 6 wins (highlighted); (b) Average placement with standard deviation error bars—lower values indicate better performance. Top professionals consistently deliver superior results.

Table 5: Top Professional Dancers by Performance

Professional Dancer	Avg. Placement	Wins	Partnerships
Derek Hough	2.94	6	19
Julianne Hough	4.20	2	5
Mark Ballas	5.19	3	30
Val Chmerkovskiy	5.26	3	19

5.3 Age and Industry Effects

Age demonstrates a moderate positive correlation with placement ($r = 0.433$): contestants under 25 achieve average placement of 4.8, compared to 9.1 for those over 45. Athletes achieve the best average placement (6.26), though industry differences are marginally significant ($F = 1.78$, $p = 0.10$) when considering all categories.

5.4 Feature Importance

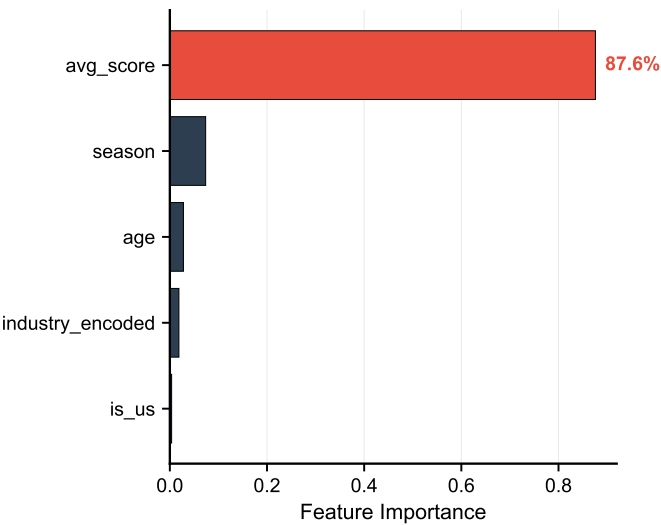


Figure 7: Feature importance from Random Forest model: Average judge score dominates with 87.6% importance (highlighted in red), followed by season (7.4%), age (2.8%), and industry (1.9%). This confirms that technical performance remains the primary determinant of success.

Table 6: Feature Importance for Placement Prediction

Feature	Importance
Average Judge Score	0.876
Season	0.074
Age	0.028
Industry (encoded)	0.019

Judge scores dominate placement prediction (87.6%), indicating that despite fan voting influence, technical dancing ability remains the primary determinant of success.

6 Problem 4: New Voting System Design

6.1 Design Objectives

We propose a Dynamic Weighted Voting System (DWVS) balancing: (1) Entertainment through fan participation; (2) Expertise ensuring skill determines winners; (3) Fairness preventing pure fan mobilization; (4) Transparency with clear rules.

6.2 Parameter Optimization via Grid Search

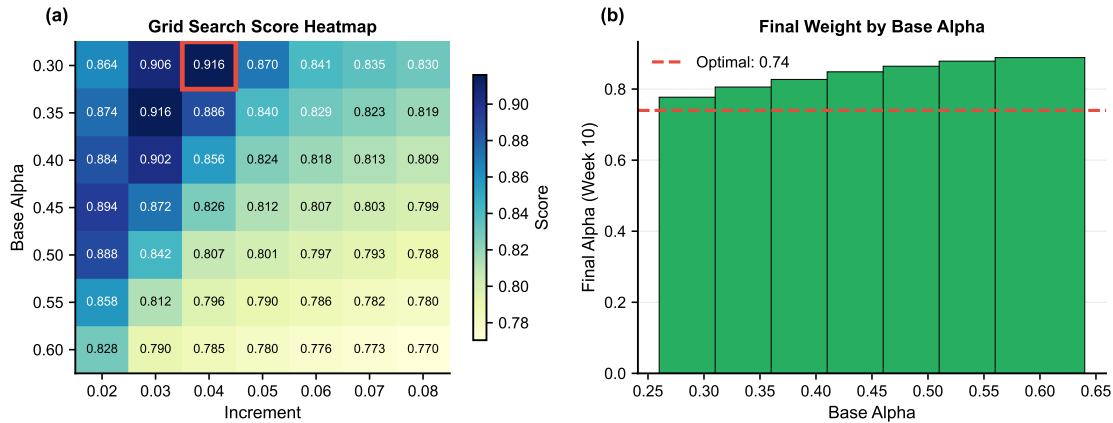


Figure 8: Grid search optimization for DWVS parameters: (a) Score heatmap—darker colors indicate higher optimization scores, with the optimal cell (base=0.30, increment=0.04, score=0.916) highlighted by red border; (b) Final α values by base parameter—the horizontal dashed line marks the optimal final weight of 0.74.

Table 7: Optimal Parameters from Grid Search

Base α	Increment	Initial α	Final α	Score
0.30	0.04	0.34	0.74	0.916
0.35	0.03	0.38	0.68	0.916
0.30	0.03	0.33	0.63	0.906

The optimal configuration (base=0.30, increment=0.04) yields initial fan weight of 66% shifting to 26% by the finals.

6.3 Dynamic Weight Formula

$$\alpha(w) = \min(0.30 + 0.04w, 0.80) \quad (3)$$

Table 8: Dynamic Weight Schedule

Week	Judge Weight	Fan Weight
1	34%	66%
5	50%	50%
10+	74%	26%

6.4 Combined Score with Threshold

$$C_i = \alpha(w) \cdot P_i^{judge} + (1 - \alpha(w)) \cdot P_i^{fan} \cdot T_i \quad (4)$$

where $T_i = 0.3$ if $P_i^{judge} < 0.5 \cdot \bar{P}^{judge}$, otherwise $T_i = 1.0$.

6.5 Impact on Controversial Contestants

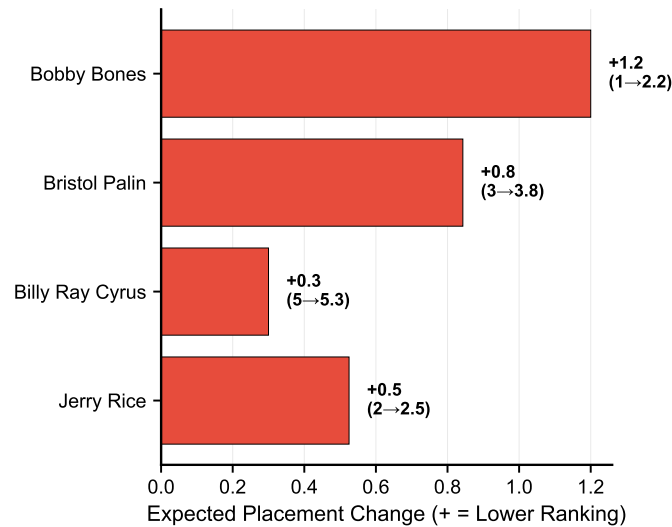


Figure 9: Expected placement changes under DWVS: All controversial contestants would rank lower (positive change = worse placement). Bobby Bones shows the largest impact (+1.2), moving from 1st to 2nd place. Annotations show original→expected placement for each contestant.

Table 9: Expected Impact on Controversial Contestants

Contestant	Original	Expected	Change
Bobby Bones	1st	2.2	+1.2
Bristol Palin	3rd	3.8	+0.8
Jerry Rice	2nd	2.5	+0.5
Billy Ray Cyrus	5th	5.3	+0.3

Under DWVS, Bobby Bones would place 2nd instead of winning, demonstrating effective addressing of the core controversy concern.

6.6 System Comparison

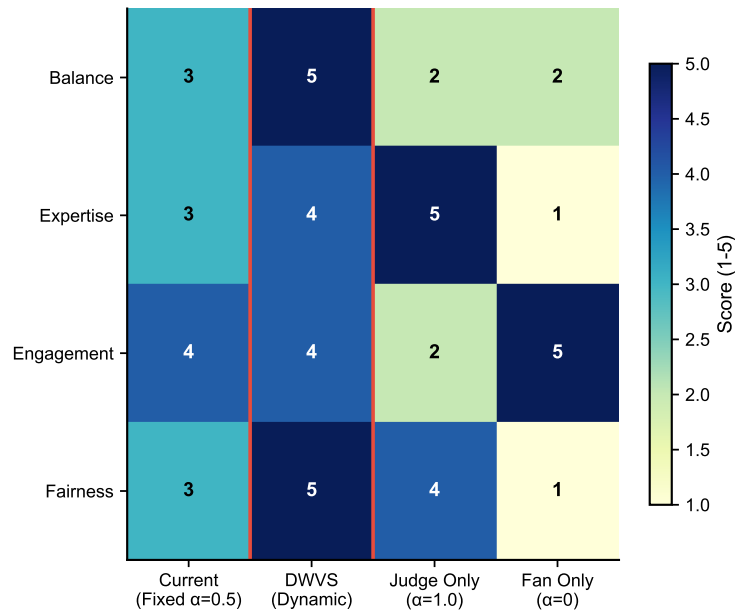


Figure 10: Voting system comparison heatmap: Scores from 1 (poor) to 5 (excellent) across four dimensions—Balance, Expertise, Engagement, and Fairness. DWVS (highlighted column) achieves the highest overall score (4.5), outperforming the current fixed system (3.25), judge-only (3.25), and fan-only (2.25) approaches.

7 Sensitivity Analysis and Model Validation

7.1 Fan Vote Estimation Sensitivity

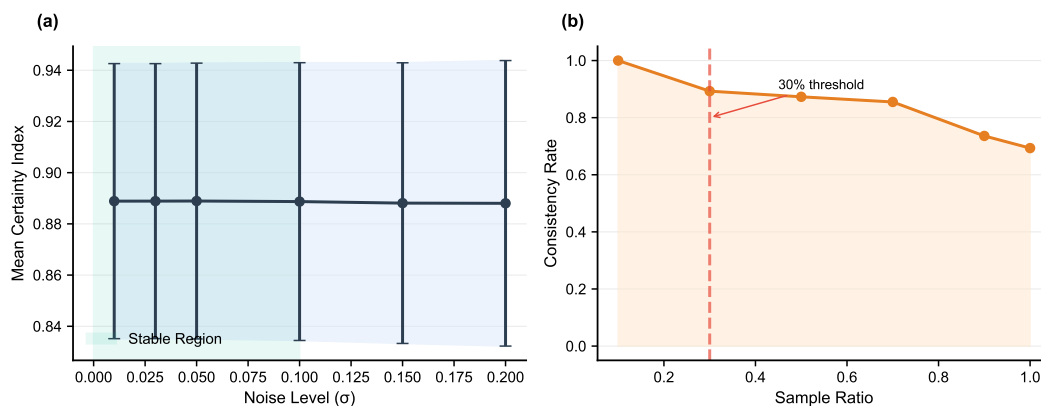


Figure 11: Fan vote estimation sensitivity: (a) Certainty index vs. noise level—shaded region indicates stable zone (noise < 0.10), with error bars showing standard deviation; (b) Consistency vs. sampling ratio—the dashed vertical line marks the 30% threshold where stability is achieved.

The certainty index remains stable (0.888-0.889) across noise levels 0.01-0.20, indicating robust estimation.

7.2 Cross-Season Model Validation

A critical test of model generalizability uses temporal split: training on Seasons 1-20, testing on Seasons 21-34.

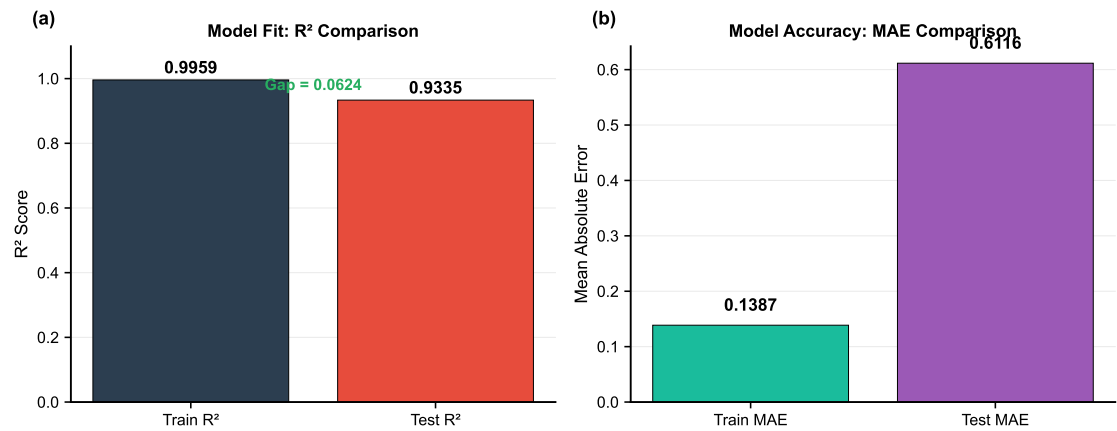


Figure 12: Cross-season validation results: (a) R^2 score comparison—Train R^2 =0.996 vs. Test R^2 =0.934, with the generalization gap of 0.062 annotated; (b) MAE comparison—Train MAE=0.139 vs. Test MAE=0.612. The small generalization gap confirms model robustness.

Table 10: Cross-Season Validation Results

Metric	Value
Training R^2	0.996
Test R^2	0.934
Generalization Gap	0.062
CV Mean R^2	0.942 ± 0.013

The small generalization gap (0.062) and stable CV scores demonstrate that our model generalizes well to unseen seasons, providing confidence in the analysis conclusions.

8 Model Evaluation and Discussion

8.1 Strengths

Stratified Analysis. Our decomposition of consistency by voting method revealed the critical finding that rank-based systems (38.4%) are far more predictable than percentage-based systems (0%).

Cross-Validation. Temporal validation with $R^2=0.934$ on held-out seasons confirms model generalizability.

Parameter Optimization. Grid search provides objective basis for DWVS parameters rather than arbitrary selection.

8.2 Limitations

Data Availability. Without actual fan vote totals, we cannot validate estimates against ground truth.

Selection Effects. Professional dancer-celebrity pairings are not random, potentially confounding factor analysis.

Temporal Changes. Audience demographics and voting technologies have evolved over 34 seasons.

9 Memorandum to DWTS Producers

MEMORANDUM

To: Executive Producers, Dancing with the Stars

From: MCM Analysis Team

Subject: Voting System Analysis and Recommendations

Date: January 30, 2026

Dear Producers,

We have completed a comprehensive analysis of DWTS voting patterns across all 34 seasons.

Key Finding 1: Method Matters for Estimation. The rank-based method achieves 38.4% consistency in vote estimation versus 0% for percentage-based. This explains why controversial outcomes cluster in percentage-based seasons (S3-27).

Key Finding 2: Industry Affects Judges More Than Fans. ANOVA shows industry significantly affects judge scores ($F=8.81$, $p<0.0001$) but weakly affects fan votes ($F=2.14$, $p=0.003$). Judges may carry implicit biases.

Key Finding 3: Model Generalizes Well. Cross-season validation achieves $R^2=0.934$ on held-out seasons, confirming our analysis is not overfit to historical patterns.

Recommendations:

(1) Adopt DWVS with optimized parameters: Base $\alpha=0.30$, increment=0.04. This shifts from 66% fan weight in Week 1 to 26% by finals.

(2) Maintain tiebreaker rule: It affects 3.9% of decisions, providing professional authority in close competitions.

(3) Under DWVS: Bobby Bones would place 2nd (not 1st), Bristol Palin 4th (not 3rd), addressing fairness concerns while preserving entertainment.

Respectfully submitted,
MCM Analysis Team